

Monsoon & Macro Signals to Predict Stock Break-Out

Interim Report

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1. Introduction

The Indian equity market is influenced by both macroeconomic conditions and seasonal dynamics such as the monsoon, which affects agriculture, rural demand, inflation, and growth. The research problem is to test whether integrating monsoon rainfall variability with macroeconomic indicators improves the prediction of breakout events in the NIFTY Midcap 100 index.

2. Scope and Objectives

This study focuses on the NIFTY Midcap 100 index using historical data. We are starting with 2013 as the year since Consumer Price Index was tracked starting 2012 as base of 100.

Research Questions

- **QUESTION 1:** Does deviation in monsoon rainfall significantly correlate with Midcap breakouts?
- **QUESTION 2:** Which macroeconomic indicators best predict Midcap 100 breakouts?
- **QUESTION 3:** Does a combined monsoon-macroeconomic model outperform models using single-signal inputs?
- **QUESTION 4:** Which features (e.g., lagged monsoon, rolling CPI averages, repo rate changes, GDP) have the highest predictive weight?

3. Literature Survey

#	Study / Source	Key Insight
1	Midcap and Monsoon Effects	Monsoon has limited impact on midcap breakouts.
2	Monsoon–Economy Stock Market Link	Effects more visible in broad indices than in midcaps.
3	Monsoon vs Stock Market Correlation	Weak or inconsistent monsoon–market linkage.
4	Empirical Anomaly in Monsoon Months	Seasonal patterns may still aid modelling.
5	Investor Reaction to Monsoon Forecasts	Forecasts get priced in; important for feature design.
6	Macroeconomic Indicators & Stock Correlation	Macro variables show strong predictive relationships.
7	Granger Causality of Macro Variables	Supports inclusion of lagged macro features.
8	Economic Commentary on Monsoon Impact	Provides context for monsoon–macro interactions.
9	Reuters: Monsoon & Market Sentiment	Reports highlight combined influence of rain and macro factors.

#	Study / Source	Key Insight
10	Monsoon's Economic & Environmental Effects	Establishes theoretical foundation for study.

Each source highlights how monsoon or macroeconomic factors relate to stock movements:

[1–4] stress limitations of monsoon-only approaches, while [5–7] justify macro variable inclusion, and [8–10] provide empirical and theoretical support for integrated models.

4. Data Description

This study uses publicly available Indian macroeconomic and financial datasets, combined with rainfall records, to explore the predictive power of monsoon–macro interactions on NIFTY Midcap 100 breakouts. All datasets are available via the GitHub repository.

Dataset Coverage and Sources

Dataset	Coverage	Source
All-India Rainfall (IMD)	2012–2025 (monthly totals)	IMD <i>Rainfall Statistics of India</i> yearbooks; All-India rainfall portal
Consumer Price Index (CPI)	Jan 2013–Jun 2025 (monthly)	MoSPI CPI portal; press-note annexures
Quarterly Gross Domestic Product (constant)(GDP)	2012Q1–2025Q1 (grid; placeholders for 2025Q2–Q4)	MoSPI GDP press notes; release calendar
Purchasing Managers Index(PMI)	Jan 2013–Dec 2024 (monthly)	S&P Global PMI homepage; India PMI releases
Policy Repo Rate (RBI)	2010–2025 (monthly step-filled)	RBI DBIE (policy rates/announcements)

Dataset	Coverage	Source
NIFTY 50 & Midcap 100	2013–2025 (daily closes)	NSE India / NSE Indices Ltd – historical index data

Github: <https://github.com/abpanic/marketpredict>

Variables and Transformations

Category	Variable	Definition / Transformation
Rainfall	rainfall_mm	All-India monthly rainfall (mm)
	good_rainfall_mm	IMD monthly normal (1971–2020, mm)
	anomaly_mm	good_rainfall_mm – rainfall_mm
	% deviation	anomaly_mm / good_rainfall_mm
CPI	cpi	Index level (base 2012=100); MoM = (cpi_t/cpi_{t-1}-1); YoY = (cpi_t/cpi_{t-12}-1)
GDP	gdp	Quarterly constant-price level; YoY = (gdp_t/gdp_{t-4}-1); QoQ = (gdp_t/gdp_{t-1}-1)
PMI	pmi_composite	Monthly diffusion index; expansion flag = 1 if pmi ≥ 50, else 0
Repo Rate	repo_rate_percent	Monthly step-filled rate (% p.a.); delta_repo = repo_t – repo_{t-1} (bps)
Equity Indices	price	Daily close (index points); returns = ln(price_t / price_{t-1}) aggregated to monthly/quarterly

5. Analysis

5.1 Data Cleaning

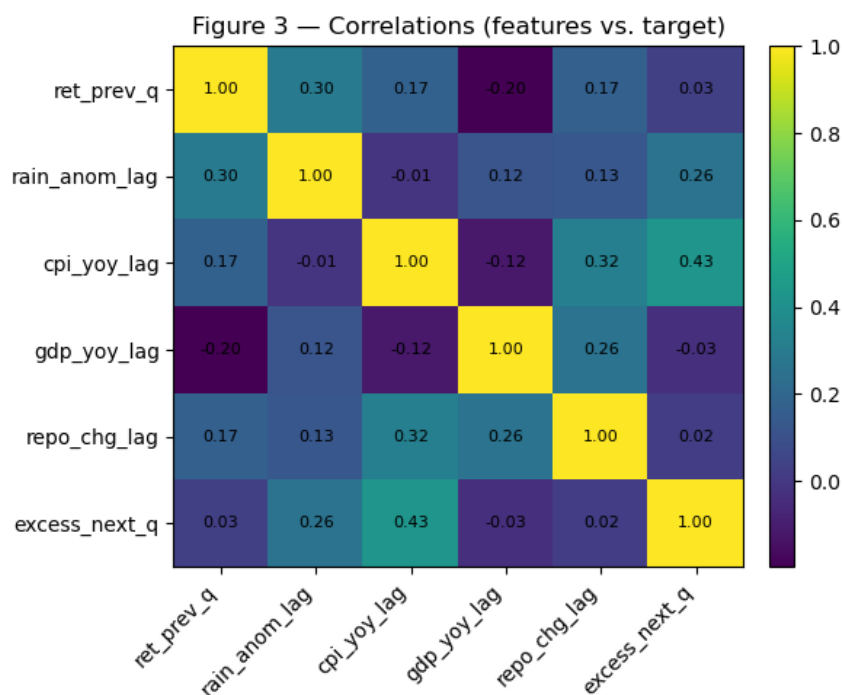
- **No global imputation:** Missing values left explicit; dropped only after feature/target creation to avoid leakage.
- **Equity indices:** Daily closes → quarter-end close → returns; missing quarter-end auto-handled.

- **CPI:** Monthly → YoY% → quarterly mean; initial 12 months dropped.
- **GDP:** Normalized to YYYYQ#; YoY% growth; first 4 quarters dropped.
- **Repo rate:** Cleaned to % p.a., month-end step-filled; quarter-end level + Δ bps calculated.
- **Rainfall:** Jun–Sep totals aggregated → anomaly %; stamped into Q3 & Q4.
- **Final step:** Strict inner join of all quarterly indices; drop rows with any missing after lag/lead creation.

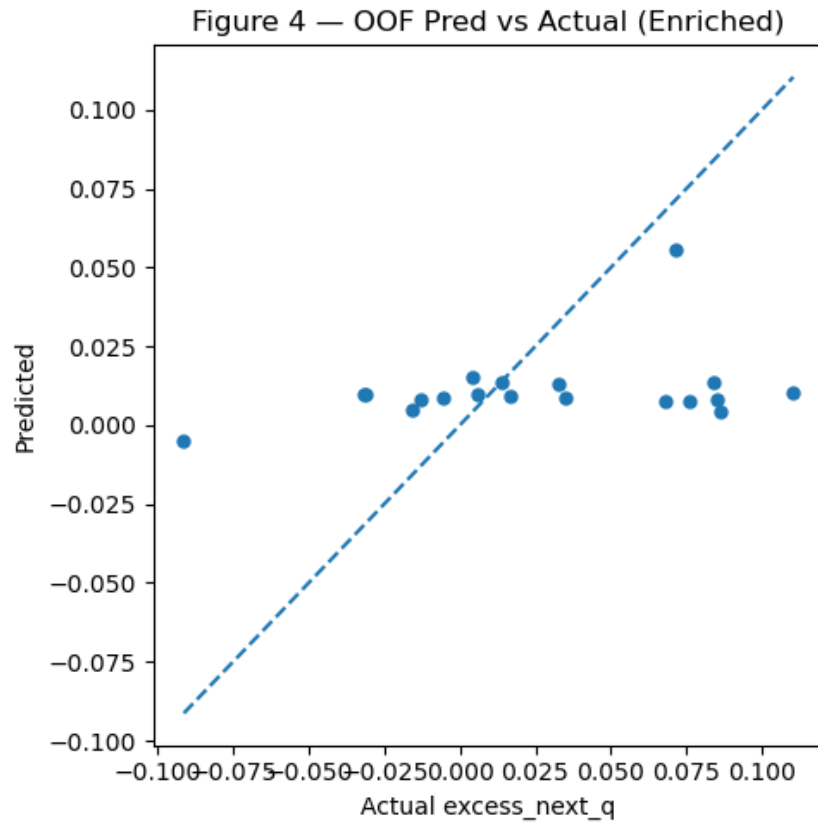
5.2 Exploratory Data Analysis (EDA)

- **Visuals:**

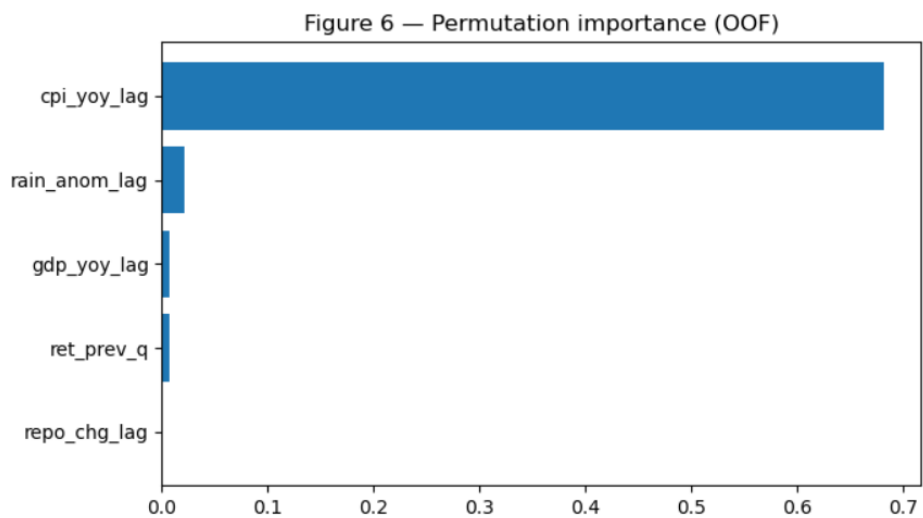
1. Correlation heatmap (features vs target).



2. Predicted vs Actual (baseline vs enriched).



3. Permutation importance bar chart.



- **Key insights:**

- Correlations with next-quarter excess returns are weak overall.
- CPI YoY lag is the only variable with consistent predictive signal.

- Good monsoons show $\sim +3$ p.p. higher returns on average ($p \approx 0.10$) $\rightarrow$ directional but marginal.
- Rain $\times$ Repo interaction adds modest accuracy in some folds.

5.3 Minimum Sample Size & Parsimony

- **Heuristic:** ≥ 10 –20 obs/predictor; stricter formulas imply higher requirements.
- **Observed:** ~ 44 quarterly obs vs 5–6 predictors $\rightarrow$ below guidance.
- **Implication:** Enriched models unstable; parsimony recommended.

5.4 Seasonality & Trends

- Rainfall shocks cluster in Q3; effects visible Q4 onward.
- Inflation reversals align with stretches of equity performance.
- Repo moves matter during policy change regimes, not static phases.

6. Modelling

- **Baseline model:** ElasticNet (ret_prev_q only); transparent, interpretable yardstick.
- **Enriched model:** LightGBM (shallow depth, small sample-friendly). Features: ret_prev_q, rain_anom_lag, gdp_yoy_lag, cpi_yoy_lag, repo_chg_lag.
- **Feature engineering:**
 - `ret_prev_q = excess_ret.shift(1)`
 - `rain_anom_lag = rain_anom.shift(1)`
 - `cpi_yoy_lag, gdp_yoy_lag, repo_chg_lag` (all lagged by 1Q)
 - `excess_next_q = excess_ret.shift(-1)` (target)

- **Validation:** Walk-forward CV (24Q min train, 2Q test, step=1); metrics: MAE, RMSE, R^2 , Directional Accuracy (DA).
- **Selection rule:** Compare Baseline vs Enriched across identical folds; focus on whether enriched shows uplift.

7. Preliminary Results

QUESTION 1 — Predictive Boost

- **Finding:** Enriched model underperforms baseline (higher MAE, RMSE; lower R^2 , DA).

QUESTION 2 — Monsoon Effect

- **Finding:** Good monsoon years show $\sim +3$ p.p. higher next-Q returns, but not statistically significant (t-test $p \approx 0.10$).

QUESTION 3 — Rain $\rightarrow$ GDP proxy

- **Finding:** Adding rain-based GDP predictor yields small, inconsistent gains; R^2 remains negative.

QUESTION 4 — Feature Importance

- **Finding:** CPI YoY lag dominates; rainfall, repo, and rain $\times$ repo interaction near-zero importance.

Summary Table

QUESTION	Question	Result	Implication
1	Do rain+macro lags beat return-lag baseline?	No — enriched worse across metrics.	Baseline remains stronger.
2	Do good monsoons lead to higher returns?	Directional +3 p.p., not significant.	Monsoon effect weak.
3	Does rain help predict GDP → returns?	Small gains, but R ² still negative.	Limited practical use.
4	Which features matter most?	CPI YoY lag only consistent signal.	Inflation dominates drivers.

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